Bug Report Classifier - Manual

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IMPORTANT

You MUST use git to clone the repository and run the code, otherwise you will not be able to replicate the results. Do NOT download the code as a zip file from GitHub, as this will not include the necessary LFS files which contain the pre-trained models.

```
git lfs install
git clone https://github.com/jaxkdev/ise_coursework
```

Note: cloning will take as long as it takes to download 2.5GB of data, so please be patient.

Please read the `requirements.pdf` file before running anything to ensure you have the necessary hardware and software requirements. **As well as a correct virtual environment setup with the required dependencies installed.**

Manual

Algorithm Training and Evaluation

To fully train and evaluate all algorithms on all projects from scratch, you can run the following command:

```
(venv) python analysis.py --cached_models=False
```

Omitting the `--cached_models=False` flag will use the cached pre-trained models instead of training from scratch, which will be much faster but will also omit training times.

Dataset Evaluation and Preprocessing

To view statistics used in the coursework and to verify datasets used, you can run the following command:

```
(venv) python src/dataset.py
```

CLI Tool Usage

The product (CLI Tool) can be used to make predictions on new bug reports using the algorithms. To use the CLI tool, run the following command:

```
(venv) python tool.py --file=path/to/bug_report.txt
```

Where `path/to/bug_report.txt` is the path to a text file containing the bug report you want to classify. The tool will output the predicted class for the bug report using the final algorithm.

To select a different algorithm, you can use the `--algorithm` flag followed by the name of the algorithm you want to use (e.g., `baseline`, `improved_1`, etc.). For example:

```
(venv) python tool.py --file=path/to/bug_report.txt --algorithm=baseline
(venv) python tool.py --file=path/to/bug_report.txt --algorithm=improved_3
```